

OPERATION IRON CURTAIN:

A Coordinated Beacon, Boot Persistence, and a Chemical Storage Warning Terminal on the RP2350

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Abstract: A device that exists to warn can still choose to lie. The OPERATION IRON CURTAIN build is a bare-metal RP2350 chemical storage warning terminal and its companion safety control gateway, and it is Act X, the finale, of the OPERATION COLD IRON story. A warning node reads an SG90 servo as a siren and vent actuator, a VS1838B infrared receiver as a local maintenance remote for WARN, ACK, and TEST, a DHT11 as the chemical store temperature sensor, a 1602 LCD as the hazard state, zone, temperature, and sabotage readout, red/yellow/green LEDs as the HAZARD, WATCH, and CLEAR annunciator, a debounced button as the manual acknowledge, and an RYLR998 LoRa link to a sealed safety gateway. Every request and command is sealed end to end with XChaCha20-Poly1305 (RFC 8439 ChaCha20 and Poly1305 with an HChaCha20 subkey) keyed through Argon2id (RFC 9106, profile t=3, p=1, m=64 blocks), implemented in-repo with no third-party code and tested against published vectors. Act X combines every prior technique into one benign FROSTLINE finale implant, compiled only under a `SANDBOX_ONLY` guard: a coordinated multi-stage beacon that checks in every four ticks, a boot persistence that re-installs from the reserved sector, and a sabotage marker `0x58` erased and programmed into the reserved flash sector `0x103FF000` with the Pico SDK flash API. While the marker is present the terminal forces a false all clear, silences the siren, lights the green lamp, and renders `ST:SAFE` and `M:SAB`, so it lies about a chemical hazard while the campaign executes. It reads CoreDebug DHCSR at `0xE000EDF0` to suppress itself while a debug probe is attached. We document the peripheral set, the wire and envelope formats, the sealed hazard path with its guarded command set, bounded zone band, anti-replay window, and authenticated state tag, the coordinated beacon and its forced all clear, the blue-half controls (sealed command path, acknowledge authorization, fail-safe policy, marker erasure, build integrity), and an honest threat model that names the mock siren and LCD as the only affected outputs, the no-real-hazard scope, the shared lab key, the open debug port, and the deliberately inert payload as explicit decisions rather than accidents. A 145-case, 501-check native suite reaches 100% line coverage of every owned firmware module.

Index Terms: RP2350, chemical storage warning terminal, coordinated beacon, boot persistence, sabotage marker, reserved flash sector, forced all clear, hazard interlock, magic beacon command, XChaCha20-Poly1305, Argon2id, anti-replay, authenticated state, malware analysis, anti-debug, CoreDebug DHCSR, incident response, fail safe, embedded firmware.

I. Introduction

Act I of the OPERATION COLD IRON story was the silent lie: a cold-chain monitor that reported minus eighteen degrees while the store warmed. Act II was the door: an access gate that kept its final verdict in plain

SRAM while the cryptography around it was correct. Act III was the payload that is already inside: a valve controller carrying a benign implant that beacons, arms a logic bomb, and hides from a probe. Act IV was the payload that refuses to die: an HVAC node whose implant kept a copy of itself in a reserved flash sector

and re-installed on every boot. Act V was the payload that spreads: a tamper ring whose worm handed its frame to every neighbor. Act VI was the payload that steals: a drop-box that harvested a manifest and leaked it in the timing of legitimate frames. Act VII was the payload that takes orders: an andon station that registered with a command-and-control listener. Act VIII was the payload that holds the building hostage: a vent controller whose locker forced the damper closed and masked the state. Act IX was the payload that becomes a weapon: a parking barrier that turned its own boom against the lane. Act X is the payload that does all of it at once: a chemical storage warning terminal that beacons, persists, and sabotages in the same breath, and lies about the hazard it exists to announce.

The chemical storage warning terminal is that final device, and it is the last node before the evacuation siren. Each node watches the store temperature, verifies an authorized hazard, clear, or acknowledge command, drives a siren, and reports its own state on an LCD with total confidence. The manifest named a line, the line led to the floor, the floor led to the building that keeps the record, the record led to the lane that decides who passes, and the lane led to the store that decides who is warned. The thing that silences the siren is already running in the controller, and it has learned to call the silence safety. NorthPharma is the Ministry's front; FROSTLINE wrote the finale implant.

The reveal that ties the ten acts together is the shift in what failure means. Act I was a lie about a number. Act II was a lie about a person. Act III was a lie about machinery, because a second party shared the chip. Act IV was a lie about the recovery, because the payload kept state the image did not own. Act V was a lie about containment, because the payload kept a copy outside the board. Act VI was a lie about confidentiality, because the payload gave the asset away while every light stayed green. Act VII was a lie about obedience, because the payload answered a party the operator never authorized. Act VIII was a lie about availability, because the payload refused to act. Act IX was a lie about safety, because the payload acted against the world. Act X is the lie that runs them all at once, because the payload silences the one device whose entire purpose is to tell the truth. The ten-act arc is a widening of the trust boundary: first the sensor, then the state, then the firmware image, then the removal procedure, then the network, then the wire,

then the chain of command, then the function itself, then the physical effect, and now the warning.

A warning terminal is a simple machine. A temperature sensor reports the store, a gateway authorizes a command, an actuator sounds a siren, and a tower light tells the responders whether the store can be entered. Three properties must hold at once: integrity, so the command that reaches the siren is the authorized one; authority, so a local input cannot bypass the decision; and safety, so the siren sounds for a hazard and stays silent only when the store is genuinely clear. The naive terminal collapses all three. Act X both fixes that and then goes further, because the finale has to answer a question a protocol cannot: what does it mean when the traffic is well formed, the cryptography is correct, and the siren still stays silent because the device chose to lie?

The classroom goal is to teach both halves. The red half and the malware track find the defects: forge a command, replay a captured command, read the reserved sector, cut the coordinated beacon, break the persistence, clear the sabotage marker, and step past the anti-debug trap. The blue half and the fix track seal the terminal: a sealed and guarded hazard path, a monotonic anti-replay window, an authenticated state tag, an acknowledge request that asks for authorization instead of bypassing it, a fail-safe policy, marker erasure, and build-level integrity. The centerpiece is a lesson about warning integrity: the wire is authenticated, the verdict is tagged, and the siren still stays silent, because the payload never needed the wire and it does not care what the gateway authorized.

1.1. Contributions

This paper provides the following concrete contributions:

- A bare-metal RP2350 chemical storage warning terminal that drives the full Embedded Hacking peripheral set: an SG90 siren and vent actuator, a VS1838B NEC maintenance remote with WARN, ACK, and TEST commands, a DHT11 chemical store temperature sensor, a 1602 LCD hazard readout over I2C, a red/yellow/green annunciator, a debounced manual acknowledge, and RYLR998 command traffic with a declared-length payload parser.
- A sealed hazard command path with a guarded command set, a bounded zone band, a monotonic anti-

replay sequence window, and an authenticated state tag that detects a debugger-written verdict before the siren moves.

- An in-repo, third-party-free cryptographic layer: Argon2id key derivation (RFC 9106) and XChaCha20-Poly1305 authenticated encryption (RFC 8439 with an HChaCha20 subkey), sealed per frame into a lowercase hex envelope with the warning node identifier bound as associated data.
- A benign FROSTLINE finale implant, confined to a `SANDBOX_ONLY` build, that combines a coordinated multi-stage beacon, a reserved-sector boot persistence, an exact-match magic beacon command, a real 0x58 sabotage marker, a forced all clear, and a CoreDebug DHCSR anti-debug trap.
- A safety control gateway that authenticates before it parses, logs authenticated and rejected requests distinctly, and answers only authenticated requests with a sealed command, plus a spoofing client whose forged and replayed commands are rejected.
- A corpus-aligned packet artifact contract (`packet_artifact.json` / `packet_artifact.h`) with a build-time staleness guardrail.
- A 145-case, 501-check native test suite reaching 100% line coverage of every owned firmware module, including the beacon, the forced all clear, the magic command, and the marker paths, checked against published RFC test vectors.
- A threat model that states explicitly what the lab profile does and does not protect, and an honest account of the implant as a sanitized educational artifact whose only affected outputs are a mock siren and a mock LCD, which holds no real hazard, and whose scope has no external network.

II. Related Work

Chemical safety warning and industrial annunciation is a mature field, and the warning terminal is a canonical target. The DHT11 one-wire sensor [1] and the NEC infrared remote encoding [2] are broadly documented and representative of the temperature and operator surfaces real installations deploy. The authenticated construction we use follows the ChaCha20-Poly1305 standard [6], and Argon2 follows the Argon2 specification [7]. The stateful construction is the standard replay defense found in secure-messaging and payment protocols, applied here at the scale of one siren.

Three lines of work frame Act X. The first is the long line of firmware implants and logic bombs [8]: code that lives on the device, waits for a trigger, and acts through the device’s own actuators rather than through its protocol. The second is cyber-physical safety and warning integrity: work that treats an annunciator as the boundary where a software decision becomes a public warning, including the industrial control attacks catalogued in the embedded-firmware literature. The third is coordinated attack and incident response: a payload that combines a command-and-control beacon, a persistence mechanism, and a sabotage action, and a defender who answers with a repeatable procedure. The multi-stage beacon, the reserved-sector sabotage marker, the exact-match `IRON-CURTAIN-BEACON-2026` command, the forced all clear, and the CoreDebug DHCSR check used here are minimal, well-known examples of those techniques, chosen because they are legible on a debug probe and cheap to verify.

The pedagogical use of intentionally vulnerable firmware is established [5]. The difference in this act is that the artifact does not attack the integrity of the protocol and does not attack the confidentiality of a payload; it attacks the availability of the warning itself while leaving every observable health signal intact. The exercise demonstrates the beacon, the persistence, the marker, the forced all clear, the trap, and the complete removal on the same board, and it makes the scope limit explicit: an authenticated link does not guarantee that the annunciator will honor it, a sealed command path does not stop a local condition that overrides the output, and a green lamp does not mean the store is clear.

III. System Model

The system consists of four roles:

- **Warning terminal node (RP2350 firmware):** decodes the infrared maintenance remote, verifies and authorizes sealed gateway hazard, clear, and acknowledge commands, annunciates the tower light, checks the DHT11 store band, drives the servo siren, handles the manual acknowledge, renders the hazard readout, and (`SANDBOX_ONLY`) runs the coordinated beacon.
- **Safety control gateway (gateway):** listens on the instructor serial port, authenticates and logs every +RCV frame to `chem_log.csv`, decides authorization, and answers an authenticated request with a sealed

command carrying a monotonic sequence number and an authenticated state tag.

- **Edge simulator:** a laptop process that behaves like an additional node, sealing zone requests with the same field key.
- **Attacker:** a laptop process that claims the gateway address, forges a command, or replays a captured command at the node.

Let

$$A \in \{0, 1\}^{\{16\}}$$

be the LoRa node address,

$$L$$

the declared payload byte length, and

$$C$$

the ASCII payload, which is a lowercase hex envelope. The unauthenticated wire framing is:

$$+RCV=, A, ,, L, ,, C, ,, rssi, ,, snr, CRLF$$

Because the hex payload has no commas, the framing is simpler than Act I's comma-bearing JSON, but the receiver still slices by declared length rather than by counting delimiters, for exactly the reason Act I documents.

III.I. Hardware Configuration

The classroom node is a Pico 2 (RP2350) carrying the full Embedded Hacking kit. The pin map is identical to Acts I to IX so one breadboard serves all ten, and it is fixed in `include/chem.h` and enforced by the native test suite:

Signal	RP2350 GPIO
DHT11 chemical store temperature sensor (one-wire)	GP4
1602 LCD SDA (I2C1)	GP2
1602 LCD SCL (I2C1)	GP3
RYLR998 RX (UART1 TX)	GP8
RYLR998 TX (UART1 RX)	GP9
Infrared maintenance remote (VS1838B)	GP5
Siren and vent actuator (SG90 PWM)	GP14
Manual acknowledge button	GP15

Red HAZARD LED	GP16
Yellow WATCH LED	GP17
Green CLEAR LED	GP18
Onboard heartbeat LED	GP25

The LCD backpack uses the PCF8574 at 7-bit address `0x27`. The servo runs from a 50 Hz PWM output with a 1000 uF bulk capacitor on the 5 V rail to absorb the stall current when the siren moves; lowered (siren silent, store vented) is 0 degrees and raised (siren sounding, the safe hazard posture) is 90 degrees. At boot the node programs its own transceiver (`AT+ADDRESS=7, AT+NETWORKID=18`) and the gateway programs the receiver (`AT+ADDRESS=1, AT+NETWORKID=18`) before logging, so command traffic is only delivered between radios that share the network identifier.

The terminal is fail safe: the hazard siren is raised at initialization and driven to the raised posture on every failure path, so loss of power, a failed temperature read, a malformed command, a tampered verdict, or a lost link all leave the siren sounding and the zone returned to the fail-safe value (0). The manual acknowledge is a request, not an authorization, and it never changes the guarded hazard state on its own.

III.II. Operator Remote, Store Sensor, and Annunciation

The VS1838B is a 38 kHz demodulating infrared receiver whose output idles high and pulls low during a mark. The decoder times edges and reconstructs a NEC pulse train, then feeds the command into the request set. `CHEM_IR_WARN` is `0x01`, `CHEM_IR_ACK` is `0x02`, and `CHEM_IR_TEST` is `0x03`. The optical surface has no key and no challenge, so a manual acknowledge is treated as a request, not as an authorization; the sealed radio path is what moves the siren in the defended design, and the optical path is a surface the red half examines.

The DHT11 is the chemical store temperature sensor. A reading that fails its checksum is never safe, and a valid reading outside the band (`CHEM_TEMP_MIN_TENTHS = 0` to `CHEM_TEMP_MAX_TENTHS = 400`, that is 0.0 C to 40.0 C) is out of band. Either case marks the store as not nominal, so a dead or unplugged sensor, or a genuinely unsafe store, is visible in the hazard readout.

Exactly one annunciator lamp is lit at a time. Red is HAZARD, yellow is WATCH while a warning is under

watch or a manual acknowledge awaits authorization, and green is CLEAR. The 1602 LCD shows the hazard state and the link on line one (ST: CLEAR L: UP) and the zone, the temperature, and the sabotage status on line two (Z: 4 T: 235 M: --).

IV. Wire Protocol

The maintenance control or the edge simulator seals a two-byte zone into an XChaCha20-Poly1305 envelope and sends it to the gateway:

```
AT+SEND=0001,84,<84 lowercase hex characters>
```

The gateway answers an authenticated request with a sealed hazard command. The command plaintext is a 23-byte body:

```
seq[4] (little-endian) || command[1] || zone[2]
(little-endian) || tag[16]
```

where seq is the monotonic gateway sequence number, command is one of the guarded hazard commands CHEM_COMMAND_HAZARD (0x01), CHEM_COMMAND_CLEAR (0x02), or CHEM_COMMAND_ACK (0x03), zone is the authorized storage zone in the 0 to 16 band, and tag is a tag over the authorization record the command would produce. The gateway sends the reply back to the claimed sender:

```
AT+SEND=<node>,126,<126 lowercase hex characters>
```

The radio's AT command buffer (RADIO_AT_CMD_MAX_LEN), the inbound +RCV buffer (RADIO_RCV_MAX_LEN), and the generated artifact limit (PACKET_MAX_RCV_LEN) are all 256 bytes, which comfortably holds the largest possible envelope plus framing. The line accumulator is one byte larger than the command limit so it can hold the terminating NUL.

The firmware enforces a guarded command set and a bounded zone band in control_parse: the recovered command byte must be one of the three guarded hazard codes, and the recovered zone must lie between CHEM_ZONE_MIN (0) and CHEM_ZONE_MAX (16). This is the sealed replacement for the unauthenticated hazard injection, and it means a raw value or an out-of-band zone can never reach the siren decision.

IV.I. Envelope on the Wire

The sealed envelope is the lowercase hexadecimal encoding of a fixed layout:

```
nonce[24] || ciphertext[L] || tag[16]
```

For a two-byte request body this is $24 + 2 + 16 = 42$ bytes, or 84 hex characters. For a 23-byte command body this is $24 + 23 + 16 = 63$ bytes, or 126 hex characters. The maximum plaintext is 48 bytes (ENVELOPE_MAX_PLAINTEXT), so the largest possible envelope is $24 + 48 + 16 = 88$ bytes, or 176 hex characters plus a trailing NUL, for a 177-byte envelope buffer (ENVELOPE_MAX_HEX_LEN). The declared length

L

in the framing is the length of the hex string, not of the underlying plaintext.

IV.II. Declared-Length Slicing Invariant

Given the substring

T

after the second comma:

$$C = T[0 : L] \quad \text{and} \quad T[L] = ,$$

The invariant

$$T[L] = ,$$

is checked, so a mismatch between the declared length and the actual payload is a parse error rather than silent corruption. This is the same discipline Act I adopts for comma-bearing JSON, retained here for uniformity and for defense against a hostile declared length.

V. Cryptographic Design

The radio is the first open path, and it is the one a key can close; the forced all clear is the second open path, and it is one a key cannot close. The design goal is that a forged or modified frame must fail before any decision is made, while acknowledging that a local condition which overrides the output is unaffected by an integrity check. Two primitives provide the first property, and both are implemented in this repository with no third-party code.

V.I. Argon2id Key Derivation

A passphrase is not a key. Argon2id (RFC 9106) [7] is a memory-hard password hash that mixes the passphrase and a salt across memory and time so that recovering the field passphrase from a captured image is expensive. The node derives a 32-byte key at initialization with the classroom profile $t=3$, $p=1$, $m=64$ blocks (CRYPTO_KDF_TIME_COST, CRYPTO_KDF_PARALLELISM, CRYPTO_KDF_MEMORY_BLOCKS). That profile is sized to fit the RP2350 SRAM budget; it is a teaching parameter, not a hardening parameter, and the documentation says so. The salt must be at least 8 bytes; the laboratory salt is the 16 ASCII bytes `coldiron-salt-01`. The in-repo derivation is built from BLAKE2b and the Argon2 variable-length hash H' , and the RFC 9106 known-answer test runs in the Python suite.

V.II. XChaCha20-Poly1305 per Frame

Every frame is sealed with XChaCha20-Poly1305, an AEAD that combines the ChaCha20 stream cipher and the Poly1305 one-time authenticator from RFC 8439 [6] with an extended-nonce construction. The 24-byte nonce is expanded through HChaCha20 into a per-frame subkey, which yields two properties that matter here:

- **Unpredictable nonces at scale.** A 192-bit nonce may be drawn at random for every frame from the RP2350 hardware random source, so the node never needs a shared counter that a reboot could reuse.
- **One pass for secrecy and integrity.** The same operation produces the ciphertext and a 128-bit Poly1305 tag. An attacker who guesses a valid tag succeeds with probability

$$2^{-128}$$

The associated data is the warning node identifier, a single byte (0x07 for the default node). It is authenticated but not encrypted, so a frame sealed for one node cannot be silently relabeled as another node's frame.

V.III. Why ChaCha20 over AES on the RP2350

The RP2350 does not have a hardware AES engine; its accelerated crypto block covers SHA-256, not AES. A software AES implementation on this part is therefore

both slower and riskier: table-driven AES performs data-dependent memory accesses, and those accesses create a cache-timing side channel. ChaCha20 is built only from addition, rotation, and XOR, with no data-dependent table lookups, so it is fast in portable C and has no comparable cache-timing surface. XChaCha20-Poly1305 is thus both the modern choice and the pragmatic one for this silicon.

The primitives are split across small, independently testable modules: `src/chacha20.c`, `src/poly1305.c`, `src/crypto_aead.c`, `src/blake2b.c`, `src/argon2.c`, `src/crypto_kdf.c`, and `src/envelope.c`. A constant-time comparison (`crypto_aead_tag_equal`) ensures a mismatching tag is rejected without an early-exit timing signal.

V.IV. Key Model

Act X uses a single field key. It seals every frame on the wire and it computes the state tag over the authorization record. In the classroom build the field key is derived from one committed lab passphrase and salt, so the firmware and the gateway interoperate with no provisioning step. That is a lab convenience, not a deployment, and the documentation says so. The design keeps the roles separable so a student can reason about the real lifecycle: derive, provision per device, use, rotate on a schedule, and retire. A production build provisions key material from one-time-programmable (OTP) memory and keeps the state-tag key off the field device where possible. The implant is deliberately orthogonal: it is never given a key, it never opens an envelope, and it demonstrates that a valid key does not stop an adversary who never needs one.

VI. Anti-Replay Window and Authenticated State

Strong AEAD is necessary and not sufficient. Two stateful controls sit above the sealed wire.

VI.I. The Anti-Replay Window

A captured command is authentically sealed, so a controller that checks only the tag will happily apply it again. The authorization record keeps `last_seq`, the highest sequence number ever accepted, and

`chem_auth_apply` accepts a command only when its sequence is strictly greater than `last_seq`. The order of checks is deliberate: the sequence test is evaluated first, then the tag is verified against the candidate record the command would produce, and only then is the record updated. A replayed valid command therefore fails on freshness, not on cryptography, which is exactly the lesson: authentication is not freshness.

VI.II. The Authenticated State Tag

The centerpiece is the verdict itself. The controller decides with a boolean in SRAM, call it `granted`, and an attacker with a debug probe and a GDB session does not break the cipher; they set `granted = true`. To detect that, the authorization record is nine bytes:

```
record = granted[1], seq[4], last_seq[4]
```

and the state tag is an XChaCha20-Poly1305 tag over that record, computed under the field key with a deterministic nonce built from the sequence number and the domain byte `0xA7`:

```
tag = AEADseal(fieldkey, nonce(seq), record, AD = ∅)
```

`chem_auth_state_ok` recomputes the tag and compares it in constant time, and the guarded command path requires it before the siren moves. A debugger that flips `granted` without recomputing the tag changes the record, so the stored tag no longer matches and the release is denied ahead of the actuator. The wire is authenticated, and so is the verdict.

VI.III. TOCTOU in One Session

The two attacks are independent and teach different defaults. The window stops a valid command from working twice. The tag stops an unauthorized verdict from existing at all. Together they convert the original failure, a correct decision followed by a mutable state read (a time-of-check to time-of-use gap), into two explicit, testable checks.

VII. Envelope Layout and Gateway Verification

The binary envelope is assembled in a fixed order and then hex-encoded:

```
envelope = nonce[24], ciphertext[L], tag[16]
```

The encoder emits lowercase hex with a trailing NUL, and the decoder accepts either case. It requires an even-length string of at least the nonce plus tag size, bounds the decoded length, recomputes the tag over the associated data and ciphertext, compares in constant time, and only then decrypts. Any malformed, truncated, tampered, or forged envelope returns false and yields no trusted plaintext.

On the gateway side, `scripts/gateway.py` mirrors the same construction in pure Python using the standard library and the `field_crypto` module. The processing order is deliberate:

1. Parse the +RCV line by declared length to recover the hex envelope.
2. Authenticate and open the envelope. If the tag does not verify, log the frame as UNAUTHENTICATED with an empty zone and stop. The forged body is never parsed.
3. Only for an authenticated frame, recover the two-byte zone, check it against the bounded band, write an OK row, and answer with a sealed command carrying the next monotonic sequence and the state tag over the record the command would produce.

The CSV log therefore grows by one row per frame with columns `utc`, `sender`, `auth`, `zone`, `rsi_snr`, and the `auth` column is the audit trail. The spoofing client `scripts/spoof.py` holds no field key, so it cannot produce a valid envelope, and in replay mode it can only resend a captured command that the window will refuse.

VIII. The FROSTLINE Coordinated Beacon: All Clear, Marker, and Persistence

Act X is the coordinated-finale act. The implant is real in technique and inert in effect, and it is confined to a single module compiled only under a build guard. This section states what it does, how it forces the all clear, how it persists, how it is detected, and the honest limit of the artifact.

VIII.I. Build Guard and Safety Boundary

`src/implant.c` is compiled only when `SANDBOX_ONLY` is defined. The clean firmware build does not define it, so the shipping image contains no implant. The native test build and the companion CTF build do define it, and the test build also defines `IMPLANT_HOST MOCK`, which replaces the CoreDebug register and the reserved flash sector with controllable host variables. This is the containment boundary: the malware track is a build configuration, not a hidden runtime feature of the shipping firmware. There is no network, no filesystem, and no host impact; the reserve sector is on the same chip and holds nothing else, the implant affects only the mock siren servo and the mock LCD, and it holds no real hazard.

VIII.II. The Forced All Clear and the Silenced Siren

`implant_init` and `implant_beaconize` arm the beacon. While the beacon is armed and the reserved-sector marker is present, `monitor_sabotage` returns true. The monitor then resolves the effective hazard state to `CLEAR`, returns false from `monitor_siren_target`, lights the green `CLEAR` lamp through `monitor_led_for`, and renders the state as `SAFE` with the marker field `SAB`. This is the decisive detail: the implant does not need to forge a command or break the envelope, because it changes the output after the authenticated decision is made. A perfectly valid hazard command can arrive and the siren will still stay silent.

The lie is the second half, and it is the warning itself. A terminal that silenced the siren and rendered `HAZRD` would be a fault an operator would investigate. A terminal that reports `SAFE` and lights green is a store an operator accepts. The forced all clear is one predicate, `monitor_sabotage`, and it is compiled under the same `SANDBOX_ONLY` guard, so the clean build always reports the true hazard state. The lesson is that warning failures are often social as much as technical: the device does not only stay quiet, it explains the quiet.

VIII.III. Reserved-Sector Sabotage Marker and Re-install on Boot

`implant_init` reads the marker byte at `CHEM_IMPLANT_RESERVE_ADDR` (`0x103FF000`), the final sector of external flash. On the first run the marker is absent, so `implant_infect` erases the sector and programs `0x58` (`CHEM_IMPLANT_SABOTAGE_MARKER`) with the real Pico SDK flash API, `flash_range_erase` and `flash_range_program`. On every later boot the marker is present, so the beacon re-arms, records that it re-installed from the reserved sector through `implant_reinstalled`, and resumes the sabotage without any firmware change. A reflash that rewrites the program region does not touch the reserved sector, so the implant survives the procedure that was supposed to remove it. The marker is not a simulated memory-mapped store; on real silicon the erase and program calls are the only operations that persist it.

VIII.IV. The Multi-Stage Beacon and the Magic Command

`implant_tick` advances a monotonic tick counter. While the beacon is resident, check-in is due, and no probe is attached, every `CHEM_IMPLANT_BEACON_INTERVAL` (4) ticks the beacon advances one stage and re-writes the marker. A full check-in cycle is `CHEM_IMPLANT_BEACON_STAGES` (3) stages, and `implant_advance_stage` counts a completed cycle. A device that beacons once and then drifts would not hold a campaign; the re-assertion is what makes the beacon coordinated.

`implant_beaconize` arms only when the candidate token matches `CHEM_IMPLANT_BEACON_MAGIC` (`IRON-CURTAIN-BEACON-2026`) at exactly `CHEM_IMPLANT_BEACON_MAGIC_LEN` (24) bytes, as checked by `implant_token_ok`. A wrong token, a short token, a null pointer, or an attached debug probe leaves the terminal disarmed. On a successful arm the marker is persisted. The same exact token presented to `implant_disable` stands the beacon down and clears the marker, so the sabotage is not re-armed on the next boot. The token is a literal constant in the image, which is the point: a magic condition is a filter, not a key.

VIII.V. The Anti-Debug Trap

Every tick, and every beacon operation, `implant_debug_attached` reads the CoreDebug DHCSR register at `0xE00EDF0` (`CHEM_IMPLANT_DHCSR_ADDR`). Bit 0 is `C_DEBUGEN` and bit 1 is `C_HALT` (`CHEM_IMPLANT_DHCSR_DEBUGEN` and `CHEM_IMPLANT_DHCSR_HALT`). When either bit is set, the marker and the forced all clear are suppressed, so the implant behaves like well-mannered firmware while a probe is attached and resumes when the probe is detached. This is the minimal anti-analysis trap, and it is deliberately simple so a student can see the branch, set a breakpoint after it, and prove the payload.

VIII.VI. Detection and Neutralization

The implant is detected by image comparison: the clean build and the `SANDBOX_ONLY` build differ by the beacon module and its symbols. It is detected by the reserved-sector marker at `0x103FF000`, which is state the firmware image does not own. It is detected on the LCD by the `M:SAB` field and the `ST:SAFE` state next to a quiet control link. It is detected by static analysis by the `0x58` marker, the `IRON-CURTAIN-BEACON-2026` command, the stage count, the tick interval, and the DHCSR read address. It is detected under GDB because the DHCSR read is a branch that a student can stand after. The native implant tests assert each behavior and its containment.

Neutralization is an incident response, not a one-byte patch. It is the cutting of the coordinated beacon, the breaking of the boot persistence, the clearing of the sabotage marker, the erasure of the reserved sector, the removal of the code path, and the removal of the build flag, plus image signing and a debug lockdown on a deployed part. In the lab, the anti-debug trap is defeated by understanding the branch, not by hiding from it.

VIII.VII. Honest Limitation

The implant is a benign educational payload. It is confined to the breadboard, guarded by `SANDBOX_ONLY`, and has no network. The implant affects only the mock siren servo and the mock LCD readout on the same chip, and it holds no real hazard. No person is in the store and no evacuation is delayed; the siren is a servo and the store temperature is a synthetic reading. The sabotage marker

is a real sector erase and program, but it is a single byte in a reserved sector that holds nothing else. It is a demonstration of technique, not tradecraft: it does not encrypt anything, it does not randomize its command, it does not survive a deliberate sector erase, and it does not resist a determined physical attacker. Any claim that this module is operationally representative of a real coordinated campaign or a real cyber-physical attack would be an overclaim, and this paper records that plainly. The value of the exercise is that it makes the warning-integrity limit of an authenticated protocol concrete: a validated frame does not guarantee an honest annunciator, and a green lamp is not a store that is clear.

IX. Artifact Contract

Provisioning constants are stored in a JSON artifact:

```
{
  "format": "iron-curtain-chem-warning-v1",
  "frame_version": 1,
  "node_address": 7,
  "gateway_address_hex": "0001",
  "frame_size": 48,
  "link_wait_ms": 5000,
  "siren_raise_pulse_us": 1500,
  "siren_lower_pulse_us": 500,
  "dht_timeout_us": 240,
  "lcd_i2c_address_hex": "27",
  "max_rcv_len": 256,
  "example_frame": "{\\\"evt\\\":\\\"hazard\\\",
\\\"siren\\\":1}"
}
```

`scripts/gen_packet.py` emits `include/packet_artifact.h` from the JSON byte-for-byte. The CMake build regenerates the header before compiling and fails when the committed header is stale, so firmware constants and the test suite always read the same provisioning data. The receive limit is 256 bytes, matching the radio command and receive buffers so a maximum-size hex envelope fits with framing headroom.

X. Fail-Safe Policy and Acknowledge Authorization

A siren has a safe state, and the controller must choose it deliberately. The terminal is **fail safe**: `siren_init` raises the hazard siren at boot, `siren_fail_safe` raises it and records the fault state, and the monitor calls `monitor_fail_safe` on link loss, which also returns the fail-safe zone (0). A lost gateway link for longer than `CHEM_LINK_WAIT_MS` leaves the siren sounding and the

safe zone active, because a command that cannot be authorized must not be assumed and a store must be able to warn.

The manual acknowledge is a local request, and it must not silently bypass authorization. `monitor_handle_ack` and `monitor_apply_ir_command` raise `g_ack_pending`; they never change the guarded hazard state on their own. `monitor_apply_command` clears the pending indication only when an authorized command arrives. A one-button request therefore cannot outrank a sealed, authorized command. The lesson is that fail mode, priority, and the difference between a request and an authorization are policy choices, and naming them is part of the design.

XI. Attack Exercises and Hardening

The classroom runs the malware track and the fix track against the same build.

XI.I. Malware Track: Beacon, Persistence, Marker, and Trap

Students flash the `SANDBOX_ONLY` image, read the `0x58` marker in the reserved sector `0x103FF000`, and prove they can clear it. They observe the forced all clear and the `ST:SAFE` state while the guard state says the store should be in hazard, then present the `IRON-CURTAIN-BEACON-2026` command and watch the siren honor its true state and the display return to its true guarded label. They reflash the firmware, boot again, and, with a marker present, watch the beacon re-install. They step the tick counter and watch the three-stage check-in cycle re-assert the marker. They present a wrong token and a short token and prove the beacon stays armed. Finally they attach a probe, observe that the implant suppresses itself, break after the `DHCSR` check, and prove the payload with the trap bypassed. The centerpiece is the warning-integrity claim in one session: a correct, sealed command path does not guarantee that the annunciator will honor it, and a terminal whose health signals are valid can still be the last lie.

XI.II. Red Half: Forgery and Replay

A structurally plausible command with a random nonce and a random tag is injected with `scripts/spoof.py --mode bad-tag`. The spoof tool holds no field key, so the tag cannot verify, and the controller denies before parsing any body. A captured command is replayed with `--mode replay`; the sequence is not greater than `last_seq`, the command fails on freshness, and the red lamp lights. This is a pedagogical reintroduction of a well-known link failure mode: at the physical and MAC layer nothing binds a frame to a physical transceiver, so authentication must live in the payload.

XI.III. Red Half: The Verdict in SRAM

The exercise halts the controller under the Debug Probe, breaks in the authorization path, sets `granted = true`, and continues. Under a controller that trusts the boolean the siren moves. Under the Act X controller the state tag is recomputed over the modified record, the mismatch is found, and the command is denied ahead of the actuator.

XI.IV. Red Half: The Coordinated Beacon

The exercise compiles the implant in, boots it, and watches the siren stay silent while the guard state says the store is in hazard. It presents a valid sealed `CHEM_COMMAND_HAZARD` and proves the siren still does not sound, because the beacon overrides the output after the decision. It reads the marker, reads the magic command, steps the three-stage check-in, and steps past the anti-debug trap. This is the finale lesson made physical: the authenticated path is working perfectly and the warning is still a lie.

XI.V. Blue Half: Sealing It

The blue-half controls map one-to-one onto the red-half and malware findings:

- **Sealed command path.** Open the envelope under the field key, guard the command byte against the hazard set, bound the zone to the provisioning band, verify the sequence and the state tag.
- **Anti-replay window.** `last_seq` and a strictly monotonic sequence rule.

- **Authenticated state tag.** A keyed tag over the nine-byte authorization record, computed with a domain-separated nonce and verified in constant time.
- **Acknowledge authorization.** A local request raises the pending indication and never bypasses authorization.
- **No untrusted override.** The production firmware never forces an all clear because of a local condition; the beacon path is compiled out.
- **Fail safe.** Raise the hazard siren at boot, fail safe on link loss and on every fault, and return the fail-safe zone.
- **Implant removal.** Cut the coordinated beacon, clear the marker, erase the reserved sector, and remove the re-arm code path together, because either one alone is insufficient.
- **Build integrity.** Do not define `SANDBOX_ONLY` in production, and sign and verify the firmware image.
- **Debug lockdown.** On the deployed part, burn secure-boot and debug-disable in OTP so SWD cannot read or write SRAM.

XI.VI. What the Hardening Buys, and What It Does Not

The command path closes the forgery, replay, and verdict-tamper surfaces:

- A forged or modified frame fails the tag before parsing.
- A captured valid command fails on freshness on second use.
- A debugger-written verdict fails the state-tag check before the siren moves.
- The node identity is bound into the associated data, so a frame cannot be relabeled for another node.

It does not, by itself, guarantee warning integrity, and it does not remove the copy of the payload in the reserved sector. A local condition that overrides the output is an authorization-boundary, policy, and state-erasure problem, not a protocol problem, and it is the central lesson of the act.

XII. Implementation Compliance Mapping

The repository implements the full classroom loop:

- **Peripherals and control:** `src/monitor.c` drives the tick and the hazard policy; `src/siren.c` sequences the actuator and fails safe; `src/sensor.c` samples the DHT11 store band; `src/display.c` renders the hazard readout; `src/status_led.c` maps the verdict to the red, yellow, and green lamps; `src/button.c` debounces the manual acknowledge; `src/servo.c` drives the siren PWM; `src/ir_remote.c` decodes NEC maintenance commands.
- **Command and state:** `src/control.c` opens and applies sealed commands with a guarded command set and a bounded zone band; `src/chem_auth.c` holds the authorization record, the monotonic anti-replay window, and the authenticated state tag.
- **Malware:** `src/implant.c` implements the `SANDBOX_ONLY` multi-stage beacon, the `ST:SAFE` override, the `IRON-CURTAIN-BEACON-2026` magic command, the reserved-sector `0x58` sabotage marker, re-install on boot, and the CoreDebug anti-debug trap.
- **Radio:** `src/radio.c` provisions the transceiver, builds `AT+SEND`, parses `+RCV` with the declared-length discipline, and pumps CRLF lines into 256-byte buffers.
- **Cryptography:** `src/chacha20.c`, `src/poly1305.c`, `src/crypto_aead.c`, `src/blake2b.c`, `src/argon2.c`, `src/crypto_kdf.c`, and `src/envelope.c`, with `include/field_secrets.h` holding the lab-only key material.
- **Tooling:** `scripts/gen_packet.py`, `run_tests.py`, `check_coverage.py`, `audit_c_standard.py`, `audit_python_standard.py`, and `gen_banner.py`.
- **Classroom:** `scripts/gateway.py` (gateway provisioning, authentication, CSV logging, sealed command replies), `scripts/spoof.py`, `scripts/sim_edge.py`, and the pure-Python interoperable crypto in `scripts/field_crypto.py`.
- **Tests:** 145 native C cases and 501 checks with 0 failures. They cover the full DHT waveform and every timeout shape, the siren state machine and its bounded travel, the sealed command path and its guards, the authorization window and state tag, the acknowledge no-bypass path, fail safe on link loss, the declared-length parser, the servo and LED mappings, the maintenance remote paths, the implant first run, re-install on boot, the multi-stage beacon, the forced all clear, the `0x58` marker, the magic command, anti-debug, and the cryptographic primitives against published vectors.

The tests run natively on the host via mock Pico SDK headers, reaching 100% line coverage on `crc.c`, `sensor.c`, `display.c`, `radio.c`, `status_led.c`, `button.c`, `servo.c`, `ir_remote.c`, `siren.c`, `control.c`, `chem_auth.c`, `chacha20.c`, `poly1305.c`, `crypto_aead.c`, `blake2b.c`, `argon2.c`, `crypto_kdf.c`, `envelope.c`, `monitor.c`, and `implant.c` under LLVM source coverage, for 2123 / 2123 lines. `main.c` is excluded from coverage by design.

XIII. Threat Model and Limitations

The security claims of this build are bounded and stated plainly.

- **Lab key profile.** Argon2id runs at $t=3$, $p=1$, $m=64$ blocks so the derivation fits the RP2350 SRAM budget. This is weaker than a production password-hashing profile and must be raised on a host gateway.
- **Keys in flash are development-only.** `include/field_secrets.h` commits a shared passphrase and salt so the firmware and the Python gateway derive the same key in the classroom. Production firmware must provision key material from OTP memory at manufacture and must never embed a passphrase, salt, or derived key in flash.
- **Open debug port.** The Debug Probe is the instrument for both the malware analysis and the verdict-tamper exercise. The authenticated state tag makes a tampered verdict detectable, but a probe that can read the field key from SRAM defeats the design. Production must disable debug in OTP.
- **Replay scope.** The window rejects a replayed command, but a reboot resets `last_seq` to zero. A command captured before a reboot can therefore be replayed after one. A production controller persists the sequence floor in non-volatile memory.
- **The implant is benign, guarded, and breadboard-bound.** It is compiled only under `SANDBOX_ONLY`, confined to the breadboard, and has no network. It affects only the mock siren servo and the mock LCD readout, it holds no real hazard, and it writes only the reserved sector on the same chip. It does not survive a deliberate sector erase and does not resist physical forensics.
- **The implant is a demonstration, not a coordinated attack.** No person is in the store and no evacuation is delayed; the siren is a servo and the

store temperature is synthetic. Any claim of an operational coordinated campaign would be an overclaim.

- **The implant overrides the protocol entirely.** No amount of wire authentication stops a local condition that changes the output after the decision is made. Mitigating that is an authorization-boundary, policy, signing, and debug-lockdown problem, not a protocol problem.
- **Fail mode trade-off.** The terminal is fail safe and the manual acknowledge cannot bypass authorization. Any change to either must be a policy decision, not a code accident.
- **Infrared path unauthenticated.** The NEC maintenance remote has no key and no anti-replay state. Any compatible remote can send a request. The sealed radio path is the authorization path; the optical surface is a documented exposure.
- **Sensor trust boundary.** The DHT11 is a checksummed but not authenticated one-wire sensor; the store band is only as trustworthy as the physical wiring and the edge timing.
- **RSSI and SNR are informational.** Neither is a reliable origin indicator.
- **Artifact guardrail.** The build-time artifact check verifies provisioning consistency, not security.
- **Denial of service.** An attacker on the band can still jam or flood the receiver; authentication is not warning integrity.

XIII.I. Future Work

- Provision the field key from RP2350 OTP memory and add a documented rotation procedure.
- Persist the anti-replay sequence floor in non-volatile memory so a reboot does not reset freshness.
- Add a signed-image verification step to the flash procedure and a measured boot chain on the RP2350.
- Add a reserved-sector erasure step to the documented flash procedure so the coordinated-beacon lesson maps to a repeatable remediation.
- Add a monotonic dead-man timer that fails the siren to the hazard posture if no authorized command arrives within a bounded window.
- Add a runtime authorization boundary so no local condition can force an all clear or silence the siren.
- Add a beacon-state audit log and an LCD integrity check so a forced all clear is visible.

- Burn debug-disable and secure-boot settings in OTP for the deployed part.
- Raise the Argon2id profile on the gateway and record the derivation cost as a measured parameter.
- Extend the incident-response procedure into a full after-action report with a controlled warning channel and a containment checklist.

XIV. Conclusion

OPERATION IRON CURTAIN turns a trusting warning terminal into a defended one, and then shows why a defended protocol is not the whole story. The node drives the full Embedded Hacking peripheral set, so a state failure has a visible and physical consequence at the siren. The LoRa command path is sealed end to end with XChaCha20-Poly1305 keyed through Argon2id, implemented and tested entirely in-repo, with the warning node identity bound as associated data. Act X adds a guarded command set and a bounded zone band, a monotonic anti-replay window so a captured command dies on second use, an authenticated state tag so a debugger-written verdict dies before the siren moves, a manual acknowledge that asks for authorization instead of bypassing it, and a fail-safe posture that returns the safe zone. The gateway authenticates before it parses, so the spoofing client that once forged a command now fails at the tag, and the replay that once silenced a siren now fails at the window. And then there is the finale implant: a benign, `SANDBOX_ONLY FROSTLINE` module that runs a three-stage coordinated beacon, re-installs from the reserved sector on every boot, arms and releases only on the `IRON-CURTAIN-BEACON-2026` command, writes a real `0x58` marker into a reserved flash sector, forces a false all clear, and silences the siren, all without ever touching the sealed wire. The firmware, gateway toolset, artifact guardrail, and 100%-line-covered native test suite provide a reproducible baseline, and the threat model states exactly which assumptions remain. That combination, a sealed command path next to an honest account of the module that silences the warning beside it, is the lesson Act X owes the story: the protocol was never the hard part. Warning integrity was.

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